THE LOCAL-LOCAL STRUCTURE DIAGRAM CAN BE CHOSEN WITH ALL THREE CORNERS AFFINE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 13522-13530 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 13532-13545, the complete author proof (12 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Affine upgrade of the collected L16061; a one-step corollary with its own distinct aggregate statement (clause (7)), counted per lane rules.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The parent lemma local-local-structure-finite-type (already collected as L16061, which supplies the diagram with geometrically integral fibres and properties (1)-(6)); Morphisms, lemma smooth-open; affineness of finite morphisms over affine targets; fibre products of affines over affines.

Source reference. Original lemma 057K: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 13432–13436. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 056G: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 13545–13545. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Assumption and notation as in Lemma 057K. In addition to properties (1) – (6) we may also arrange it so that*

(7) *S', Y', X' are affine.*

Proof. Note that if Y' is affine, then X' is affine as π is finite. Choose an affine open neighbourhood $U' \subset S'$ of s'. Choose an affine open neighbourhood $V' \subset h^{-1}(U')$ of y'. Let $W' = h(V')$. This is an open neighbourhood of s' in S', see Morphisms, Lemma 056G, contained in U'. Choose an affine open neighbourhood $U'' \subset W'$ of s'. Then $h^{-1}(U'') \cap V'$ is affine because it is equal to $U'' \times_{U'} V'$. By construction $h^{-1}(U'') \cap V' \to U''$ is a surjective smooth morphism whose fibres are (nonempty) open subschemes of geometrically integral fibres of $Y' \to S'$, and hence geometrically integral. Thus we may replace S' by U'' and Y' by $h^{-1}(U'') \cap V'$. $\square$

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.